
Confounded Oracles Misguide Biomolecular Design: Population-Structure Shortcuts in Antimicrobial Candidate Optimization

Anonymous Authors
Anonymous Affiliation
anonymous@domain.edu

Abstract

Predictive oracles are central to in-silico biomolecular design, but optimizing an oracle is only useful when its learned associations remain valid for the biological target of interest. Using a controlled semi-synthetic stress test parameterized by the resistance prevalence of 3,975 *Mycobacterium tuberculosis* records—where genomic features, peptide latent factors, and drug-target efficacy are all simulated—we investigate how population-specific shortcuts in AMR prediction propagate into antimicrobial-peptide latent-space optimization. Standard predictors retain rank discrimination ($\text{AUC} \approx 0.80$) but encode spurious lineage-associated shortcuts. CORAL strongly reduces source-target covariance discrepancy ($>99.6\%$) but leaves the shortcut intact; DANN’s adversarial training erodes transferable signal without reliably eliminating domain or shortcut information. In a latent-space design loop pairing these oracles with a Character-Level VAE, a covariate-shift control with a shared efficacy mechanism confirms that alignment is benign or actively helpful. Under mechanism shift, aligned oracles show directionally lower mean true efficacy than the unaligned source oracle, though the paired differences are not significant at the current replication scale ($n=10$ model seeds). However, under a prior-penalized constrained control preventing latent extrapolation, CORAL produces significantly poorer outcomes ($p_{\text{Holm}} = 0.008$). Representation probes show distinct average failure patterns across independently trained model seeds. Overall, this demonstrates a failure mode that can arise when biological oracles are optimized under mechanism shift.

1 Introduction

Antimicrobial resistance (AMR) is a global health crisis, demanding rapid and accurate diagnostics. Deep learning models applied to whole-genome sequencing (WGS) data have shown remarkably high predictive accuracy. However, bacterial populations are highly structured into clonal lineages, and resistance determinants are often heavily correlated with these lineages. This population-structure confounding leads to catastrophic failure when models are deployed on novel, unseen bacterial clades, as recently demonstrated in large-scale empirical studies [9] and lineage-dependency analyses [10]. This creates a particularly consequential failure mode for biomolecular design: a predictor need not merely make incorrect predictions; when used as an optimization objective, its shortcuts can determine which molecules are generated at all.

Beyond clinical diagnostics, generative antimicrobial design pipelines commonly rely on learned activity predictors or screening oracles to rank novel antimicrobial peptide (AMP) candidates [6, 14]. If those oracles inherit population-specific resistance shortcuts, this is no longer just an accuracy issue—it directly corrupts the generative design loop, causing the

pipeline to prioritize ineffective molecules. Indeed, recent work has independently recognized that optimizing learned molecular-property predictors can produce high predicted scores but poor true outcomes [12, 13].

We ask when population-structure deconfounding produces a safer biological oracle and when enforcing invariance instead removes information required for target-specific molecular optimization. Prior work has separately documented population-structure confounding in AMR prediction [9, 10] and predictor hacking in generative molecular design [12, 13]. We connect these failure modes in a controlled semi-synthetic framework.

In this paper, we construct a controlled semi-synthetic stress test parameterized by clinical resistance prevalences (3,975 isolate phenotypes) that models the separation of causal features from spurious lineage markers. We first evaluate whether adversarial deconfounding architectures, including DANNs and CORAL [1], can correct the diagnostic failure. Finding that they cannot, we extend the analysis to a generative design loop, demonstrating how this lingering confounding can misdirect downstream candidate optimization.

2 Related Work

AMR Prediction and Population Structure: Numerous studies have applied Random Forests and Deep Neural Networks to predict AMR [2]. Most are evaluated using random train/test splits, obscuring the effect of population structure. Yu et al. demonstrated across five pathogens and >24,000 genomes that population structure is a major confounder [9]. Billows et al. specifically analyzed lineage dependency in *M. tuberculosis* AMR prediction [10]. **Correcting Population Structure:** In genome-wide association studies (GWAS), linear mixed models [3, 4] are routinely used to correct for population structure. Porting these natively into deep learning remains challenging. Moyer et al. showed that adversarial invariance training can itself be counterproductive, removing useful information alongside nuisance variables [11].

Domain Adaptation: DANNs [5] and CORAL [1] aim to align feature distributions across domains. Zhao et al. showed that domain-invariant representations need not transfer under conditional shift when the source and target labeling mechanisms differ [15]; our mechanism-shift experiment instantiates this failure in an oracle-driven biomolecular optimization setting. While other approaches like IRM [7] or Group DRO [8] attempt to address distribution shifts, we focus on the widespread alignment paradigms.

Generative Antimicrobial Design and Oracle Reliability: Recent work uses VAEs and deep generative models to discover novel antimicrobials [6, 14]. These pipelines assume the downstream screening oracle accurately reflects causal biology, an assumption we directly challenge, echoing recent concerns on the limitations of in silico benchmarking [16]. Gendreau et al. simulated predictor hacking during goal-directed molecular generation [12], and Yoshizawa et al. developed reliability-aware molecular optimization to address reward hacking [13]. Our contribution connects the biological source of oracle failure (population-structure confounding) to the downstream design consequence.

3 Methodology

3.1 Data Simulation Protocol

The 3,975 BV-BRC *M. tuberculosis* records were used only to estimate isoniazid resistance prevalence. Synthetic examples were then assigned to two balanced population-structure domains (1,987 and 1,988 examples); these domains do not represent reconstructed biological *M. tuberculosis* lineages. We constructed a 50-dimensional genomic feature matrix X_{bact} according to probabilistic logic gates:

1. Causal Mechanism (x_0): $x_0 \sim \text{Bernoulli}(0.402)$. The resistance phenotype is conditionally generated as $P(y = 1|x_0 = 1) = 0.761$ and $P(y = 1|x_0 = 0) = 0.090$, yielding a marginal base rate of ≈ 0.359 .
2. Spurious Lineage Marker (x_1): In Domain 0: $P(x_1=1|y=1) = 0.99$, $P(x_1=1|y=0) = 0.01$. In Domain 1: $x_1 \sim \text{Bernoulli}(0.1)$, independent of y .
3. Background ($x_{2..49}$): Domain-specific features ($x_{2..9}$ in Domain 0, $x_{10..17}$ in Domain 1; $P = 0.9$) and global noise ($x_{18..49}$; $P = 0.1$). Full specification in Appendix C.

3.2 AMP Generative Model and Drug-Target SCM

We generated 5,000 synthetic cationic, amphipathic peptide sequences and trained a Character-Level VAE to compress them into a continuous latent space $Z_{\text{amp}} \in \mathbb{R}^{16}$. We deliberately employ this basic VAE as a minimal generative baseline to isolate oracle failure; while it lacks the capacity for diverse physicochemical outputs (e.g., favoring R/K-enriched sequences), this serves our purpose of a controlled stress test rather than state-of-the-art AMP discovery. We paired the bacterial isolates with sampled VAE latent vectors and constructed a hybrid drug-target SCM with two regimes:

Regime 1 — Covariate Shift (positive control): The causal efficacy mechanism is identical in both clades. Source and target efficacy are both $\sigma(4.0 \cdot x_0 \cdot z_0 - 1.0)$. Only the nuisance lineage distribution differs.

Regime 2 — Mechanism Shift: The target lineage has mutated such that latent activity factor A (z_0) becomes an antagonist, and only latent activity factor B (z_1) is effective. Source efficacy: $\sigma(4.0 \cdot x_0 \cdot z_0 - 1.0)$. Target efficacy: $\sigma(4.0 \cdot x_0 \cdot z_1 - 4.0 \cdot x_0 \cdot z_0 - 1.0)$.

3.3 Model Setup and Evaluation Schemes

We define the UDA protocol formally. Let $D_S = \{(x_i, y_i, c_i=0)\}$ be the labeled source and $D_T = \{(x_j, c_j=1)\}$ be the unlabeled target. The task loss uses source labels only; the DANN domain loss and CORAL alignment loss use both source and unlabeled target embeddings; target labels are revealed only at evaluation. We evaluated a Linear Model (OLS), Logistic Regression, Random Forest, a Standard NN, a DANN, and a CORAL network [1]. All diagnostic experiments use 20 random seeds.

3.4 Drug-Target Oracle Architecture

Each design oracle is a joint drug-target predictor $f_\theta : \mathbb{R}^{50} \times \mathbb{R}^{16} \rightarrow [0, 1]$ that takes the concatenation $[X_{\text{bact}}, Z_{\text{amp}}]$ as input (66 dimensions) and is trained with BCE loss against the SCM efficacy outcome. The architecture is identical to the diagnostic classifiers (2-layer MLP: $66 \rightarrow 32 \rightarrow 16$, ReLU activations, single-unit output head). For UDA oracles (DANN/CORAL), unlabeled Clade 1 drug-target pairs are used for alignment losses but their efficacy labels remain hidden during training. During optimization, we fix a target bacterium x^* from Clade 1 with $x_0^* = 1$ and perform gradient ascent on Z : $Z_{t+1} = Z_t + \eta \nabla_Z f_\theta(x^*, Z_t)$ for 100 steps (Adam, $\eta = 0.1$), maximizing the oracle’s predicted efficacy. We condition on $x_0=1$ targets because $x_0=0$ yields peptide-independent efficacy under the SCM ($\sigma(-1) \approx 0.269$ regardless of Z). All oracles share identical random Z initialization per replicate.

4 Results

4.1 Diagnostic Performance

Classical baselines (OLS, Logistic Regression, Random Forest) retained substantial rank discrimination under the population-structure shift (target AUC ≈ 0.80), although default-threshold balanced accuracy remained near chance. The Standard NN overfit to the source lineage (0.716 AUC). DANN’s alignment penalty caused catastrophic degradation (target AUC 0.546), while CORAL experienced a milder but substantial drop (0.685 AUC). Even minimal adversarial pressure ($\lambda = 0.01$) causes an immediate, severe drop in target AUC ($0.733 \rightarrow 0.555$); full diagnostic metrics are in Appendix Table 3 and the λ -sweep in Appendix Table 5.

4.2 Latent Causal Probe

To identify the exact mechanism of alignment failure, we trained linear probes on the frozen latent embeddings of the design oracles (66-dimensional input; Section 3.4) to decode both the causal resistance mechanism (x_0) and the spurious lineage marker (x_1) from target-domain representations. Probes were evaluated on a held-out 30% split of target-domain embeddings and we report mean $\pm$ std across 10 independently trained model seeds (Figure 1).

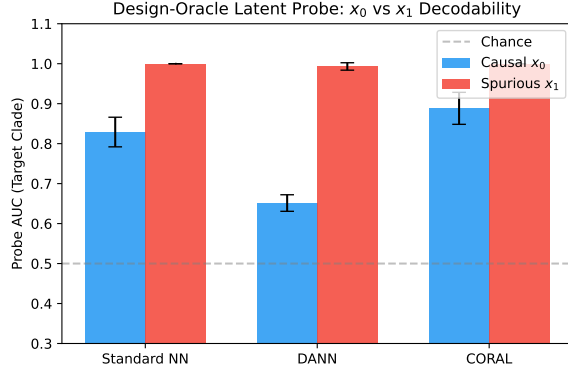


Figure 1: Design-oracle latent probe: decodability of causal (x_0) vs. spurious (x_1) features from target-domain embeddings of the 66-dimensional drug-target oracles. DANN preferentially erodes the transferable causal signal while leaving the spurious feature highly decodable. CORAL preserves both. Error bars: ± 1 std across 10 model seeds.

The Standard NN heavily encodes the spurious marker (x_1 AUC = 1.000 ± 0.000) and partially retains the causal signal (x_0 AUC = 0.829 ± 0.037). DANN preferentially erodes the transferable causal signal (x_0 drops to 0.651 ± 0.021) while leaving the spurious feature highly decodable ($x_1 = 0.993 \pm 0.009$). An independent post-hoc domain probe (logistic regression on frozen DANN embeddings; Appendix D) yields AUC = 1.000 across all 10 seeds, confirming that DANN’s representations remain perfectly domain-separable despite adversarial training. CORAL preserves both signals nearly intact ($x_0 = 0.888 \pm 0.040$, $x_1 = 1.000 \pm 0.000$) while effectively reducing covariance distance between domains ($>99.6\%$ reduction vs. Standard NN; Appendix D).

4.3 Design Loop: Mechanism Shift vs. Covariate Shift

We deployed these predictors as design oracles (Section 3.4) against a novel target-domain bacterium. Oracle A is a Target-Informed Reference with access to target labels; Oracles B, C, and D are trained in the strictly UDA setting (source labels only). To ensure proper statistical replication, we used a nested design: 10 independently trained oracle seeds $\times$ 10 matched latent initializations per seed. For each model seed we compute a seed-level mean efficacy $\bar{E}_s$, then conduct a two-sided Wilcoxon signed-rank test across the 10 model seeds with Holm correction for two comparisons within each regime (Table 1).

Table 1: Design-loop true SCM efficacy under covariate shift (shared mechanism) vs. mechanism shift. Model-seed averages (10 seeds $\times$ 10 latent starts). $\Delta \bar{E}$: paired difference vs. Standard NN at the model-seed level.

Oracle	Covariate Shift	Mechanism Shift	$\Delta \bar{E}$ (Mechanism)	p_{Holm}
A (Target-Informed)	0.979 ± 0.040	0.891 ± 0.261	—	—
B (Standard NN)	0.989 ± 0.031	0.264 ± 0.296	ref.	ref.
C (DANN)	0.900 ± 0.300	0.189 ± 0.333	-0.075	0.770
D (CORAL)	1.000 ± 0.000	0.213 ± 0.290	-0.051	0.465

Under covariate shift, where the causal mechanism is shared across domains, CORAL and the Standard NN achieve near-ceiling efficacy (1.000 ± 0.000 and 0.989 ± 0.031 , respectively), while DANN remains high on average but substantially more variable (0.900 ± 0.300). Under mechanism shift, all source-trained oracles achieve low true efficacy. The aligned methods have lower mean efficacy than the unaligned oracle (DANN: 0.189 ± 0.333 ; CORAL: 0.213 ± 0.290 ; Standard NN: 0.264 ± 0.296), but seed-level effects are variable and the paired differences are not statistically significant ($p_{\text{Holm}} = 0.770$ and 0.465 , respectively; two-sided Wilcoxon signed-rank, $n=10$ model seeds). Design-oracle probes (Figure 1) provide the stronger signal, revealing distinct failure mechanisms: DANN erodes the transferable x_0 signal (0.651 ± 0.021), whereas CORAL retains it (0.888 ± 0.040) along with the shortcut.

To disentangle oracle-reliability effects from off-manifold latent extrapolation, we repeated the design experiment under a prior penalty: $\max_Z f_\theta(x^*, Z) - \beta \|Z\|_2^2$ with $\beta = 0.1$. This

constrained evaluation supplements the main unconstrained design loop by evaluating 5 novel target bacteria per model seed (with 10 paired latent starts per target), yielding a $10 \times 5 \times 10$ nested evaluation design. This keeps latent norms near the VAE prior region ($\|Z\| \approx 1.9$ for Standard NN and CORAL, vs. $\|Z\| \approx 17\text{--}25$ unconstrained).

Table 2: Prior-penalized design-loop true SCM efficacy ($\beta = 0.1$) under covariate shift vs. mechanism shift (averaged across 5 targets per model seed). $\Delta\bar{E}$: paired difference vs. Standard NN (Holm-corrected for two comparisons per regime).

Oracle	Covariate Shift		Mechanism Shift	
	Mean Efficacy	p_{Holm}	Mean Efficacy	p_{Holm}
Target-Informed	0.976 ± 0.021	—	0.924 ± 0.165	—
Standard NN	0.865 ± 0.215	ref.	0.098 ± 0.187	ref.
DANN	0.892 ± 0.298	0.106	0.198 ± 0.380	0.770
CORAL	0.981 ± 0.029	0.027	0.004 ± 0.007	0.008

Under constrained optimization in the covariate-shift regime, prior-penalized CORAL achieves significantly higher efficacy than the Standard NN ($p_{\text{Holm}} = 0.027$, mean $\Delta E = +0.116$, 95% CI $[-0.029, 0.261]$). Under mechanism shift with the exact same optimizer, CORAL achieves significantly lower true efficacy than the Standard NN (9/10 seeds negative, $p_{\text{Holm}} = 0.008$, mean $\Delta E = -0.094$, 95% CI $[-0.230, 0.043]$). The DANN result remains non-significant due to high variance from outlier seeds. A projection-based constraint ($\|Z\| \leq 4.0$) produces a similar pattern but without significance (Appendix details available in supplementary code). All constrained candidates remain R/K-enriched regardless of oracle, confirming that decoded peptide diversity is limited by the synthetic VAE training data. This clean interaction (difference-of-differences signed-rank test across model seeds, $p = 0.0039$) isolates mechanism shift as the cause of the CORAL degradation.

5 Discussion and Conclusion

In this controlled semi-synthetic stress test, population-structure shortcuts degrade source-to-target oracle reliability and propagate into latent-space peptide optimization. The positive control confirms that when source and target share the same efficacy mechanism, CORAL and the Standard NN achieve near-ceiling efficacy; under mechanism shift, all source-trained methods deteriorate.

The representation probes provide the strongest result. CORAL nearly eliminates covariance discrepancy ($>99.6\%$ reduction; Appendix D) while the shortcut remains perfectly decodable (x_1 AUC = 1.000). An independent domain probe shows DANN’s representations remain perfectly domain-separable (probe AUC = 1.000; Appendix D), while DANN erodes the causal signal (x_0 AUC: $0.829 \rightarrow 0.651$) without eliminating shortcut information (x_1 AUC = 0.993). Under prior-penalized optimization preventing latent extrapolation, the CORAL oracle reliably degrades design ($p_{\text{Holm}} = 0.008$, 9/10 seeds negative), confirming this failure persists near the VAE manifold. The core failure is not merely poor domain adaptation, but rather a fundamental inability to handle conditional shift: enforcing invariance across domains whose underlying causal mechanism has changed [15].

Limitations. This is a controlled semi-synthetic stress test, not peptide-engineering validation. Pathogen features, domains, and drug-target efficacy are simulated; BV-BRC data provide only resistance prevalence. The mechanism-shift setting intentionally violates shared-labeling assumptions [15]. Even with prior-penalized optimization ($\|Z\| \approx 2$), decoded candidates converge to R/K-rich sequences (Appendix E); the VAE lacks capacity for diverse physicochemical outputs. We interpret the design experiment as an oracle-reliability benchmark that demonstrates a specific risk pathway when biological oracles are optimized under mechanism shift, rather than a proof that current AMR pipelines are clinically unsafe. Real genomic features, realistic AMP data, and experimental validation remain necessary. Code and simulator specification are provided at <https://anonymous.4open.science/r/AMR-519B>; training details are in Appendix B.

References

- [1] Sun, B., & Saenko, K. (2016). Deep coral: Correlation alignment for deep domain adaptation. In *Computer Vision–ECCV 2016 Workshops* (pp. 443-450).
- [2] Moradigaravand, D., et al. (2018). Prediction of antimicrobial resistance in *Escherichia coli* from large-scale pan-genome data. *PLoS computational biology*, 14(12), e1006258.
- [3] Earle, S. G., et al. (2016). Identifying lineage effects when controlling for population structure improves power in bacterial association studies. *Nature microbiology*, 1(5), 1-6.
- [4] Lees, J. A., et al. (2018). pyseer: a comprehensive tool for microbial pangenome-wide association studies. *Bioinformatics*, 34(24), 4310-4312.
- [5] Ganin, Y., et al. (2016). Domain-adversarial training of neural networks. *The journal of machine learning research*, 17(59), 1-35.
- [6] Das, P., et al. (2021). Accelerated antimicrobial discovery via deep generative models and molecular dynamics simulations. *Nature Biomedical Engineering*, 5(6), 613-623.
- [7] Arjovsky, M., et al. (2019). Invariant risk minimization. *arXiv preprint arXiv:1907.02893*.
- [8] Sagawa, S., et al. (2019). Distributionally robust neural networks. *arXiv preprint arXiv:1911.08731*.
- [9] Yu, Y., Wheeler, N. E., & Barquist, L. (2025). Biased sampling driven by bacterial population structure confounds machine learning prediction of antimicrobial resistance. *PLOS Biology*, 23(12), e3003539.
- [10] Billows, N., et al. (2023). Feature weighted models to address lineage dependency in drug-resistance prediction from *Mycobacterium tuberculosis* genome sequences. *Bioinformatics*, 39(7), btad428.
- [11] Moyer, D., et al. (2018). Invariant representations without adversarial training. In *Advances in Neural Information Processing Systems* (pp. 9084-9093).
- [12] Gendreau, M., et al. (2023). Molecular assays simulator to unravel predictors hacking in goal-directed molecular generations. *Journal of Chemical Information and Modeling*, 63(13), 3983–3998.
- [13] Yoshizawa, T., et al. (2025). A data-driven generative strategy to avoid reward hacking in multi-objective molecular design. *Nature Communications*, 16, 2409.
- [14] Szymczak, P., et al. (2023). HydrAMP: a deep generative model for antimicrobial peptide discovery. *Nature Communications*, 14, 1453.
- [15] Zhao, H., et al. (2019). On learning invariant representations for domain adaptation. In *Proceedings of the 36th International Conference on Machine Learning* (pp. 7523–7532). PMLR.
- [16] Surana, S., Grinsztajn, N., Atkinson, T., Duckworth, P., & Barrett, T. D. (2025). Overconfident Oracles: Limitations of In Silico Sequence Design Benchmarking. *arXiv preprint arXiv:2502.17246*.

A Diagnostic Performance

Table 3: Model Performance under Random (Scheme A) and UDA (Scheme B) Splits. Mean $\pm$ Std (20 seeds).

Model	Sch. A AUC	Sch. B AUC	Sch. B Bal. Acc.	Sch. B PR-AUC
Linear Model (OLS)	0.941 $\pm$ 0.000	0.797 $\pm$ 0.000	0.515 $\pm$ 0.000	0.626 $\pm$ 0.000
Logistic Regression	0.935 $\pm$ 0.007	0.797 $\pm$ 0.000	0.515 $\pm$ 0.000	0.627 $\pm$ 0.000
Random Forest	0.953 $\pm$ 0.007	0.805 $\pm$ 0.004	0.516 $\pm$ 0.001	0.634 $\pm$ 0.004
Standard PyTorch NN	0.941 $\pm$ 0.008	0.716 $\pm$ 0.055	0.522 $\pm$ 0.006	0.538 $\pm$ 0.047
Adversarial DANN ($\lambda = 1.0$)	0.649 $\pm$ 0.058	0.546 $\pm$ 0.074	0.508 $\pm$ 0.037	0.405 $\pm$ 0.066
Deep CORAL (Weight = 1.0)	0.925 $\pm$ 0.012	0.685 $\pm$ 0.016	0.555 $\pm$ 0.031	0.526 $\pm$ 0.015

B Architecture and Hyperparameters

All deep learning models (Standard NN, DANN, CORAL) share a common feature encoder architecture: a 2-layer Multi-Layer Perceptron (MLP) with 32 and 16 hidden units, using ReLU activations. The classification head is a single linear layer mapping from 16 to 1 dimension. For the DANN, the domain discriminator matches the classification head (a single linear layer), preceded by a Gradient Reversal Layer. Following the original method of Ganin et al. (2016), the adversarial penalty weight is annealed from 0 to λ over training using $\lambda_p = \lambda \left(\frac{2}{1 + \exp(-10 \cdot p)} - 1 \right)$ where p progresses linearly from 0 to 1. The CORAL penalty computes the squared Frobenius norm between the covariance matrices of source and target embeddings, scaled by a fixed weight of 1.0. All models were trained for 250 epochs using the Adam optimizer with a learning rate of 0.005, weight decay of 1×10^{-4} , and BCE loss.

The Character-Level VAE uses an embedding layer (21 tokens $\rightarrow$ 32-dim) followed by two 1D convolutions (64, 128 channels), projecting to a 16-dimensional latent space ($\mu, \log \sigma^2$). The decoder mirrors this with a linear layer, two transposed convolutions, and softmax output. Trained for 10 epochs (Adam, lr= 10^{-3} , $\beta_{KL} = 0.1$) on 5,000 synthetic AMP sequences (15–30 amino acids, enriched for cationic and hydrophobic residues).

Design oracles were trained independently for each of 10 model seeds. For each model seed, 10 matched latent initializations were used, with each latent start shared across all four oracle types. Hyperparameters ($\lambda = 1.0$, CORAL weight= 1.0) were fixed a priori; target labels were used only for final evaluation.

C Complete Simulator Specification

For each sample i , with assigned domain $c_i \in \{0, 1\}$:

1. $x_0 \sim \text{Bernoulli}(0.402)$.
2. $y \sim \text{Bernoulli}(0.761)$ if $x_0=1$, else $y \sim \text{Bernoulli}(0.090)$.
3. If $c=0$: $x_1 \sim \text{Bernoulli}(0.99)$ if $y=1$, else $\text{Bernoulli}(0.01)$; $x_{2..9} \sim \text{Bernoulli}(0.9)$; $P(x_{10..17}=1) = 0$.
4. If $c=1$: $x_1 \sim \text{Bernoulli}(0.1)$ (independent of y); $x_{10..17} \sim \text{Bernoulli}(0.9)$; $P(x_{2..9}=1) = 0$.
5. $x_{18..49} \sim \text{Bernoulli}(0.1)$ (global noise).

Peptide latent vectors $Z_{\text{amp}} \in \mathbb{R}^{16}$ are sampled i.i.d. from the standard normal prior; one vector is assigned per bacterial sample (3,975 pairs total).

D Alignment Metrics

To confirm that alignment objectives are actually optimized, we report:

- DANN domain-classifier accuracy (final epoch): 0.798 ± 0.240 across 10 model seeds.
- Independent DANN domain probe: A logistic regression classifier trained post-hoc on frozen DANN encoder embeddings to predict source vs. target domain achieves $\text{AUC} = 1.000 \pm 0.000$ across all 10 seeds. Despite adversarial training, the DANN representation remains perfectly domain-separable.
- CORAL covariance distance: Standard NN 0.260 ± 0.070 vs. CORAL 0.0008 ± 0.0004 (>99.6% reduction).

CORAL successfully reduces second-order covariance discrepancy without removing the shortcut. DANN’s adversarial training does not reliably eliminate domain information; the clean interpretation is that DANN substantially damages causal-feature decodability while leaving domain and shortcut features fully recoverable.

E VAE Decode Validity

For each of 10 model seeds, we decoded the optimized latent vector (first latent start) through the trained VAE decoder and report peptide-level metrics:

Table 4: Decoded peptide properties from optimized latent vectors (10 model seeds).

Oracle	Valid	Unique	Avg Length	Avg Charge	Avg Hydro.
Target-Informed	10/10	10/10	30.0	+30.0	−4.32
Standard NN	10/10	10/10	30.0	+30.0	−4.33
DANN	10/10	10/10	30.0	+30.0	−4.32
CORAL	10/10	10/10	30.0	+30.0	−4.34

All optimized latent points decode into valid amino-acid sequences. However, all decoded candidates are heavily R/K-enriched (charge $\approx +30$, hydrophobicity ≈ -4.3), reflecting the cationic bias of the synthetic training set rather than meaningful oracle-driven sequence diversity. This confirms that the design-loop analysis should be interpreted as a latent-space oracle stress test rather than a peptide-chemistry result.

F Lambda Sweep

Table 5 details the impact of the DANN adversarial weight (λ) on model performance.

Table 5: Impact of DANN Adversarial Weight (λ) on AUC. Mean $\pm$ Std (20 seeds).

λ (Max Alpha)	Scheme A AUC	Scheme B AUC
0.00	0.908 $\pm$ 0.013	0.733 $\pm$ 0.048
0.01	0.665 $\pm$ 0.059	0.555 $\pm$ 0.066
0.05	0.629 $\pm$ 0.019	0.526 $\pm$ 0.035
0.10	0.642 $\pm$ 0.035	0.534 $\pm$ 0.057
0.50	0.633 $\pm$ 0.016	0.548 $\pm$ 0.073
1.00	0.649 $\pm$ 0.058	0.546 $\pm$ 0.074

G Training Diagnostics

Figure 2 and Figure 3 present the variance diagnostics and t-SNE embeddings, respectively.

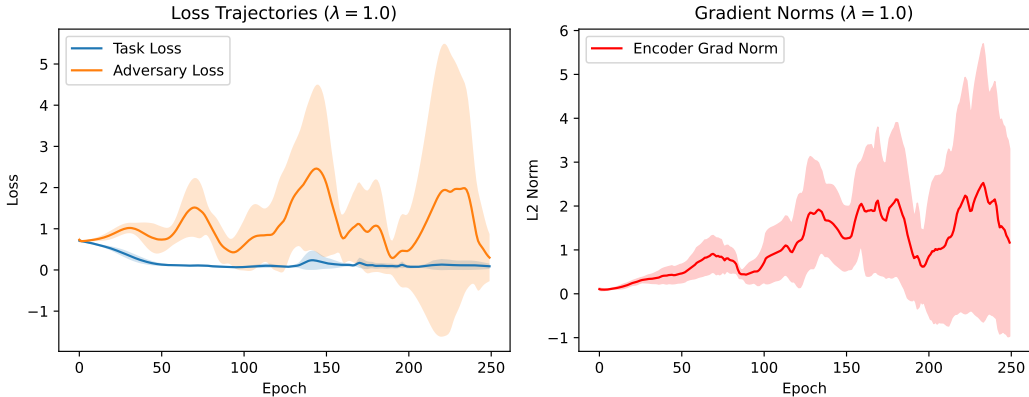


Figure 2: Variance Diagnostics: Loss trajectories and encoder gradient norms over 250 epochs for the DANN model ($\lambda = 1.0$). The adversarial penalty induces significant instability despite the recommended Ganin et al. annealing schedule.

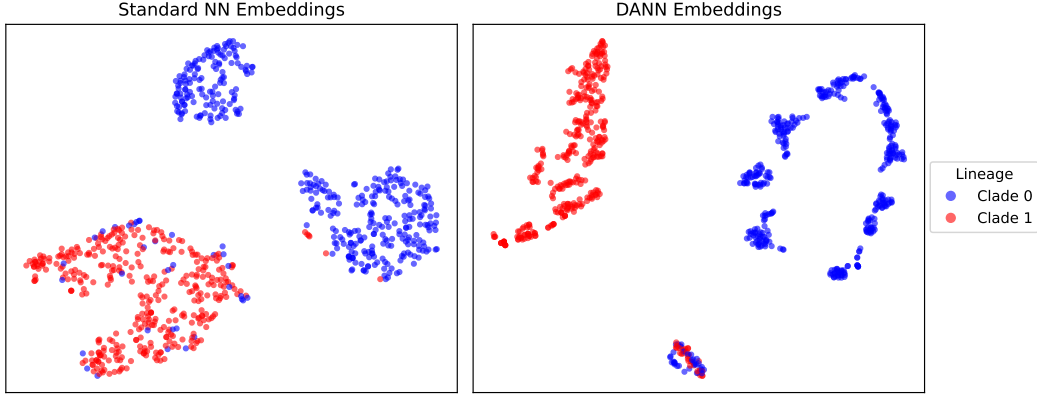


Figure 3: t-SNE visualization of the latent representations colored by domain. Domain alignment attempts to close the gap but fails to remove spurious lineage-associated separation.

H Optimization Trajectory

Figure 4 presents the unconstrained optimization trajectory under mechanism shift.

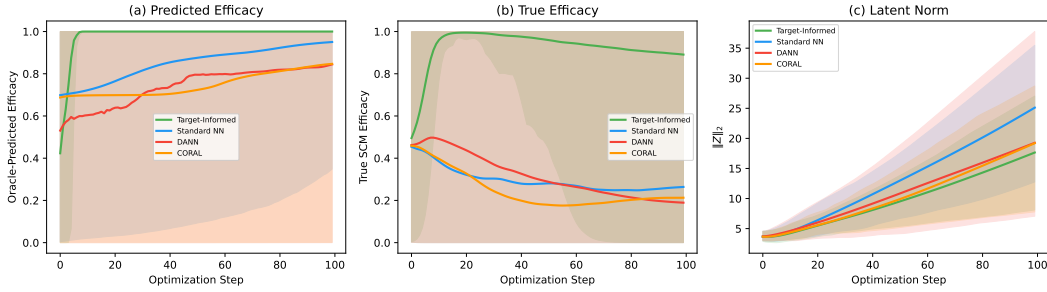


Figure 4: Optimization trajectory under mechanism shift (mean $\pm$ 95% CI across 100 replicates: 10 model seeds $\times$ 10 latent starts). (a) All oracles increase their predicted objective. (b) True SCM efficacy: only the Target-Informed oracle improves; DANN/CORAL diverge further than the Standard NN. (c) Latent norm $\|Z\|_2$ grows during optimization for all oracles.

LLM Usage Disclosure

In the preparation of this manuscript, LLMs were utilized as an assistive pair-programming tool for code generation and prose refinement. All experimental results, methodological designs, and final conclusions were directed, reviewed, and verified by the human authors, who bear full responsibility for the contents and scientific integrity of this work.